

Client Lead Management System (Mini CRM)

Software Requirements Specification

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1. Introduction

1.1 Purpose

The Client Lead Management System is a small lead-tracking application intended for one- and two-person small businesses. It captures inbound enquiries from a website contact form, tracks each lead through a simple lifecycle, and provides the business owner with a single place to view leads, notes, activity, and follow-up information.

The system is intentionally narrow in scope. Its purpose is to make it easy for the business owner to identify which leads require attention and review the most recent information associated with each lead.

1.2 Scope

In scope:

- Admin authentication.
- Lead creation, viewing, updating, searching, filtering, and deletion.
- Lead notes.
- Lead activity history.
- Public lead capture through a contact form.
- Follow-up date tracking.
- Realtime updates to an open dashboard.
- Responsive and accessible user interfaces.

Out of scope:

- Email sending.
- Multi-user accounts.
- Sales pipelines and forecasting.
- Reporting and analytics.
- Native mobile applications.
- Third-party integrations.
- File uploads.
- Bulk import/export.
- Custom fields.
- Password reset by email.
- Two-factor authentication.
- Localisation.

1.3 Users

Primary user: A single business owner (admin), assumed to be non-technical and likely to use the system on a mobile device.

Secondary user: A public visitor submitting an enquiry through the contact form. The visitor does not authenticate and interacts only with the public lead-capture interface.

1.4 Definitions and Abbreviations

Term	Definition
Admin	The authenticated business owner who manages leads.
Lead	A person or organisation represented by an enquiry or potential business opportunity.

Activity	An immutable record describing an action performed on or concerning a lead.
Note	Text attached to a lead to record additional information.
Realtime event	A server-generated event delivered to an active dashboard connection.
NFR	Non-functional requirement; a requirement describing a quality, constraint, or characteristic of the system.

1.5 References

- ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.
- World Wide Web Consortium (W3C), *Web Content Accessibility Guidelines (WCAG) 2.1*.

2. Overall Description

2.1 Product Perspective

The system consists of an authenticated administration interface for the business owner and a public contact form for visitors. The administration interface provides lead management, notes, activity history, and follow-up information. The public interface allows a visitor to submit a new enquiry.

The system is a standalone application and does not depend on third-party business integrations in version 1.

2.2 Main Functions

The system shall provide the following high-level capabilities:

- Authenticate the administrator.
- Manage leads and their attributes.
- Record notes and activity history.
- Accept public enquiries.
- Notify connected dashboards of relevant changes.
- Provide a responsive and accessible user interface.

2.3 Operating Environment

The system shall be usable through a modern web browser. The administration interface shall support mobile, tablet, and desktop screen sizes.

2.4 Constraints and Assumptions

- There is one administrator account in version 1.
- The system is designed for a small number of leads rather than large-scale CRM workloads.
- Production traffic shall use HTTPS.
- Configuration that differs between environments shall be supplied through environment-specific configuration.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide:

- An admin login interface.
- An admin dashboard for viewing and filtering leads.
- A lead-detail interface for viewing and updating a lead.
- A public contact form for submitting enquiries.

The detailed presentation requirements for these interfaces are defined in Section 4.7.

3.2 Software Interfaces

The system shall communicate between its web client and backend through the application's API. Lead, note, activity, and authentication data shall be managed by the backend and persisted by the application's data store.

3.3 Communication Interfaces

Authenticated and production communication shall use secure transport. Realtime updates shall use a one-way server-to-client communication channel.

4. Functional Requirements

4.1 Authentication

FR-AUTH-1 — The system shall authenticate the single admin using an email address and password.

FR-AUTH-2 — The system shall issue an authenticated session after a successful login, with the session remaining valid for 7 days unless invalidated earlier.

FR-AUTH-3 — The system shall provide a logout operation that invalidates the current session.

FR-AUTH-4 — The system shall provide an operation to retrieve the current admin's profile information, including identifier, email, and name.

FR-AUTH-5 — The system shall reject requests to protected resources when the session is missing, malformed, or expired.

FR-AUTH-6 — The system shall not disclose whether a given email address exists when authentication fails.

4.2 Lead Management

Lead Attributes

Attribute	Required	Constraint / Notes
Name	Yes	1–120 characters.
Email	Yes	Valid email format.
Phone	No	0–30 characters.
Company	No	0–120 characters.
Source	Yes	Contact form, referral, WhatsApp, or other.
Status	Yes	New, contacted, converted, or lost. Default: new.
Follow-up date	No	Nullable.

Last contacted at	No	Nullable; automatically set when the lead transitions to contacted unless explicitly supplied in the same update.
Created at / Updated at	Auto	Maintained by the system.

Requirements

FR-LEAD-1 — The system shall return a list of leads to the authenticated admin, with filtering by status and searching across name, email, and company.

FR-LEAD-2 — The system shall allow the admin to create a lead using the defined lead attributes.

FR-LEAD-3 — The system shall return a single lead together with its notes and activity records.

FR-LEAD-4 — The system shall allow the admin to update name, email, phone, company, source, status, and follow-up date.

FR-LEAD-5 — The system shall allow the admin to delete a lead and shall also remove that lead's notes and activities.

FR-LEAD-6 — The system shall reject writes when lead attributes violate the defined validation constraints.

FR-LEAD-7 — The system shall reject status values outside the defined status set.

FR-LEAD-8 — When a lead transitions to contacted, the system shall set the last-contacted-at value automatically unless a value is explicitly supplied as part of the same update.

FR-LEAD-9 — The system shall not return lead data to an unauthenticated caller.

FR-LEAD-10 — The system shall paginate lead-list responses using a default page size of 50 and a maximum page size of 200.

4.3 Notes

FR-NOTE-1 — The system shall allow the admin to append a note to a lead.

FR-NOTE-2 — The system shall reject a note when its body is empty or contains only whitespace.

FR-NOTE-3 — The system shall reject a note when the associated lead does not exist.

FR-NOTE-4 — The system shall return a lead's notes ordered from newest to oldest.

FR-NOTE-5 — Notes shall be immutable in version 1; the system shall not provide note-edit or note-delete operations.

4.4 Activity Log

FR-ACT-1 — The system shall return a lead's activity entries ordered from newest to oldest.

FR-ACT-2 — The system shall automatically create an activity record for every lead mutation without requiring the client to supply the activity record.

FR-ACT-3 — Activity records shall be immutable. The system shall not provide an operation to modify or individually delete an activity record, except when it is removed through cascade deletion of its parent lead.

FR-ACT-4 — Failure to create an activity record shall not prevent the primary lead mutation from succeeding.

FR-ACT-5 — The system shall present activity entries as human-readable descriptions rather than raw activity-type codes.

4.5 Public Lead Capture

FR-CAP-1 — The system shall accept an unauthenticated submission containing name, email, optional phone, optional company, source, and message.

FR-CAP-2 — On successful submission, the system shall create a lead with status new, record a lead-created activity attributed to the system, and notify connected administration dashboards of the new lead.

FR-CAP-3 — The system shall apply the same lead-validation rules to public submissions as to admin-created leads, where those rules are applicable.

FR-CAP-4 — The system shall reject or silently discard submissions identified as automated or abusive through the system's anti-abuse mechanism without revealing the internal reason for rejection.

FR-CAP-5 — The system shall limit public capture submissions to 5 per source IP address per hour.

FR-CAP-6 — The system shall store the submitted message as a note attached to the new lead rather than as a lead attribute.

FR-CAP-7 — The system shall not disclose internal identifiers or whether the submitted email address already exists in the response to a public submission.

4.6 Realtime Updates

FR-RT-1 — The system shall deliver a realtime event to connected dashboards whenever a lead is created, updated, or deleted, or when an activity record is created.

FR-RT-2 — Realtime event delivery shall be one-way from the server to the client.

FR-RT-3 — The realtime connection shall remain available during normal idle periods.

FR-RT-4 — The client shall automatically attempt to reconnect after a dropped realtime connection.

FR-RT-5 — After reconnection, the client shall reconcile changes that may have been missed while disconnected.

FR-RT-6 — The admin's own mutations shall be reflected immediately in the interface and shall not depend on successful realtime delivery.

FR-RT-7 — The client shall display a visible indication when the realtime connection is unavailable.

4.7 User Interface Requirements

FR-UI-1 — The login screen shall provide email and password fields and shall display an inline error on authentication failure without clearing the entered email address.

FR-UI-2 — The dashboard shall display leads as cards below 768px and as a table at 768px and above.

FR-UI-3 — The dashboard shall provide a status filter and text search across lead name, email, and company.

FR-UI-4 — Each lead summary shall display name, email, status, source, and follow-up date, with the follow-up date visually emphasised when it is due today or overdue.

FR-UI-5 — The lead-detail interface shall display metadata, status controls, follow-up date controls, notes, and the activity timeline in that order. Notes and activities shall be displayed newest first.

FR-UI-6 — Every user-initiated mutation shall provide immediate interface feedback. When the server rejects the operation, the interface shall display an error and restore the previous state where appropriate.

FR-UI-7 — The public capture form shall collect name, email, optional phone, optional company, source, and message. On successful submission it shall display a confirmation and clear the form.

FR-UI-8 — The interface shall support light and dark themes, allow the user to toggle between them, persist the selected theme across sessions, and use the operating-system preference as the default when no selection has been made.

FR-UI-9 — The interface shall meet WCAG 2.1 Level AA contrast requirements for supported themes.

FR-UI-10 — All interactive elements shall be operable using a keyboard and shall provide a visible focus indicator.

5. Non-Functional Requirements

5.1 Performance

NFR-PERF-1 — The frontend shall become interactive within 3 seconds on a 4G connection, excluding backend cold-start time.

NFR-PERF-2 — Authenticated API responses shall complete within 500 ms under normal operating load, excluding backend cold-start time.

NFR-PERF-3 — The system shall tolerate a backend cold start of up to 60 seconds without an unrecoverable client error and shall indicate a pending slow request after 2 seconds.

NFR-PERF-4 — Realtime events shall normally reach connected clients within 2 seconds of the triggering mutation.

NFR-PERF-5 — The dashboard shall render a dataset containing 100 leads without a noticeable reduction in usability under normal operating conditions.

NFR-PERF-6 — The initial JavaScript bundle shall not exceed 500 KB gzipped.

5.2 Security

NFR-SEC-1 — All production traffic shall use HTTPS.

NFR-SEC-2 — Passwords shall be stored using a one-way password-hashing mechanism such that the original password cannot be recovered from the stored value.

NFR-SEC-3 — Session credentials shall not be accessible to client-side JavaScript.

NFR-SEC-4 — Cross-origin access shall be restricted to an explicit allowlist; wildcard origins shall not be permitted in production.

NFR-SEC-5 — All externally supplied input shall be validated server-side before use.

NFR-SEC-6 — Database queries shall be parameterised; application code shall not construct queries through unsafe string concatenation.

NFR-SEC-7 — Production secrets shall not be committed to the source repository.

NFR-SEC-8 — Production error responses shall not expose stack traces, database errors, or internal identifiers.

NFR-SEC-9 — Passwords and other authentication secrets shall never be written to application logs.

NFR-SEC-10 — Security-related HTTP response headers shall be configured to reduce common browser-based risks, including content-type sniffing, framing, and unsafe referrer disclosure.

5.3 Usability

NFR-USE-1 — A first-time admin shall be able to log in, view leads, and change a lead's status without external instruction.

NFR-USE-2 — User-initiated mutations shall provide visible acknowledgement of receipt within 200 ms under normal client conditions.

NFR-USE-3 — Destructive actions shall require explicit user confirmation before being completed.

5.4 Reliability

NFR-REL-1 — The realtime connection shall reconnect automatically after a drop using exponential backoff capped at 30 seconds.

NFR-REL-2 — The client shall reduce accidental duplicate submissions by disabling or otherwise preventing repeated activation of a mutation while that mutation is in progress.

NFR-REL-3 — A backend restart shall not result in permanent loss or corruption of previously persisted data.

5.5 Maintainability

These requirements describe lightweight development constraints rather than runtime behaviour. They are intentionally kept here because this specification is for a small personal project.

NFR-MNT-1 — Server-side code shall be organised by responsibility, including routing, request handling, domain logic, data access, and validation.

NFR-MNT-2 — Client-side code shall be organised by responsibility, including pages, components, hooks, and API access.

NFR-MNT-3 — Environment-specific configuration shall be supplied through environment variables or equivalent external configuration.

5.6 Portability and Development

NFR-PORT-1 — The application shall run locally using documented commands after dependencies have been installed.

NFR-PORT-2 — Deployment shall not require manual modification of generated build artefacts.

5.7 Accessibility and Responsiveness

NFR-A11Y-1 — The application shall meet WCAG 2.1 Level AA requirements applicable to its supported interfaces.

NFR-A11Y-2 — All form inputs shall have an associated accessible label.

NFR-A11Y-3 — All interactive elements shall be operable using a keyboard.

NFR-A11Y-4 — Touch targets shall be at least 44 px by 44 px where practical and applicable.

NFR-RESP-1 — The dashboard shall remain usable at a viewport width of 375 px without requiring horizontal scrolling for normal content.

NFR-RESP-2 — The layout shall adapt across mobile, tablet, and desktop viewport sizes.

6. Domain Model

6.1 Entities

The main domain entities are:

- User
- Lead
- Note
- Activity

6.2 Relationships

- One User may author many Notes.
- One Lead may have many Notes.
- One Lead may have many Activities.
- One User or the system may generate many Activities.
- Deleting a Lead cascades to its Notes and Activities.

6.3 Lead Lifecycle

The defined lead statuses are:

- new
- contacted
- converted
- lost

The statuses describe the intended lifecycle of a lead. Version 1 does not enforce a restricted transition graph; the admin may change a lead between the defined statuses. When the status becomes *contacted*, the system applies the last-contacted-at behaviour specified by FR-LEAD-8.

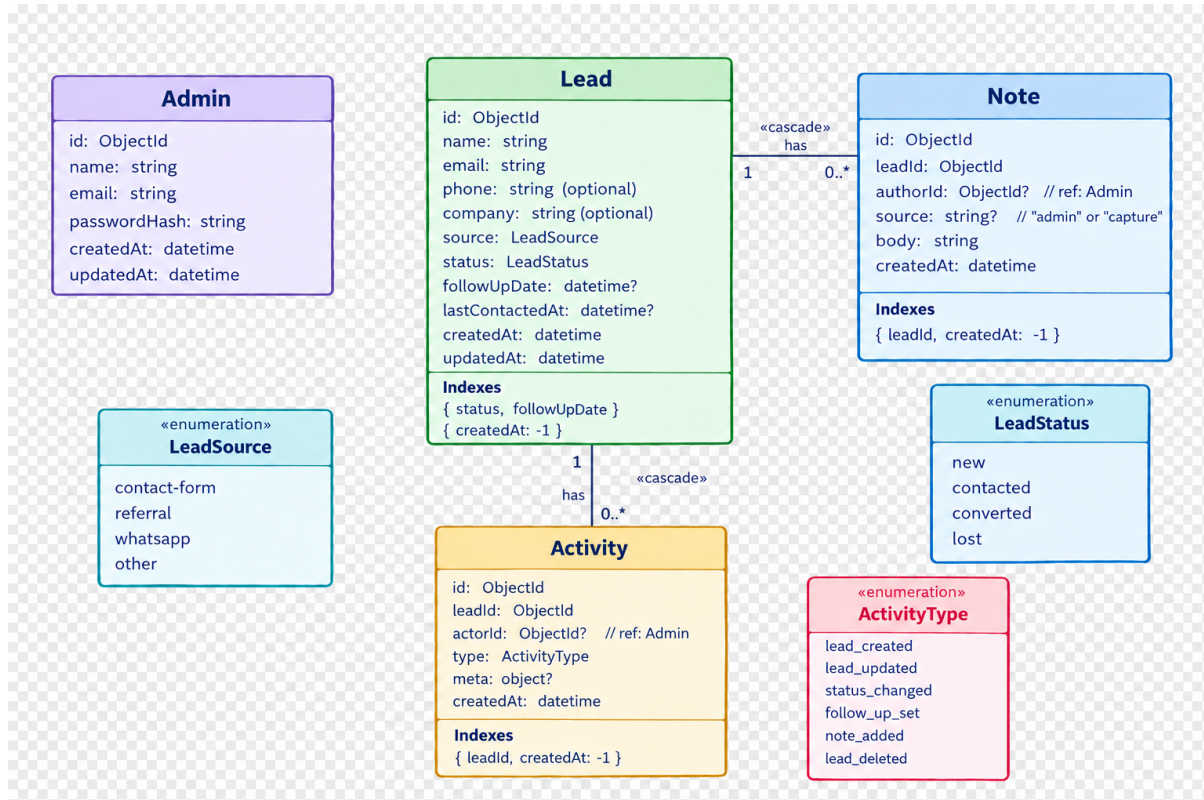


Figure 1: Entity relationship diagram for the Client Lead Management System.

7. Conclusion

This specification defines the functional, user-interface, non-functional, and domain requirements for the Client Lead Management System.